Module 2: Decide whether to simulate or not

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Description

Simulate yes or no?

Prerequisites

- Module 'Introduction to simulation'

Learning Objectives

You can:

- decide when it makes sense to use a simulation and when not.
- recognize whether a simulation study is suitable for a specific question.
- name the advantages and disadvantages of using simulation.

Time Requirement

- 20 minutes

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Overview

Why is this learning unit needed and what does it contain?

In this learning unit, you will learn how to make informed decisions about whether a simulation study is useful in a particular context or not. But why is this even important?

The world around us is becoming increasingly complex. Whether in logistics, production, healthcare or environmental analysis - decisions often have to be made in the face of uncertainties resulting from a multitude of influencing factors. In such cases, simulations can help to understand interrelationships, run through various scenarios and make well-founded decisions.

However, simulation is not always the best solution. The use of simulation studies requires time, resources and often a high level of expertise. It is therefore crucial to answer the question:
Simulate - yes or no?

In this learning unit you will find out:

- When it makes sense to use simulations.
- The advantages and disadvantages of simulations.
- How you can weigh up whether a simulation study is appropriate in a specific case.

The content is supplemented by examples, visualizations and interactive exercises to help you apply what you have learned directly.

Look forward to sharpening your analytical skills and getting to know an important problem-solving tool.

Have fun exploring, and let's go!

When to simulate?

Weighing up the costs and benefits: When is it worth using a simulation?

The decision to use a simulation is a balancing act between cost and benefit. It is important to carefully weigh up whether the use of a simulation will actually provide the necessary insights or whether simpler methods will suffice. The aim is to use simulation resources effectively to make informed decisions without incurring unnecessary costs or delays.

Although simulation is a good tool, it is not always used in reality as it is not appropriate for every problem. Factors such as the excessive complexity of the system, the availability of relevant data and the resources required for modeling and running the simulation play a decisive role.

Reasons for a simulation

There are various reasons for a simulation:

- **An investigation on a real system would be too complex, too expensive or too dangerous.** For example, an expansion of a road or road system is planned in order to solve a traffic problem. During the planning phase, however, it is not yet clear whether the traffic problem can be solved by the expansion or whether the bottleneck will simply be moved to another location. This is where simulation helps to make the right decisions. This also includes training and education of employees: in simulated environments, employees can safely gain experience and learn without endangering real resources.
- **A system does not exist.** Examples include the construction of a new cargo handling terminal in a port or the construction of a new factory. In the simulator, the planned new construction or expansion is modeled and tested under the future workload. The interaction of the various systems, vehicles and machines is examined with regard to the target values.
- **The real system cannot be observed directly.** Examples: molecular simulation, simulation of geological processes,

simulation of climate change.

- **The problem cannot be solved satisfactorily using analytical and numerical optimization methods.** You can find an example in the drop-down box below.
- **There are a large number of variants or options for action.** Examining the effects of these variants cannot be done efficiently “by hand”. Once the simulation model has been set up, a large number of options can be simulated with little effort.
- **The aim is to illustrate complex interrelationships.** Simulation tools are also used for visualization in order to make the effects of various influences visible and understandable (e.g. congestion at a junction depending on the number of lanes or traffic signal control).

In some cases, the effort required to create an accurate simulation may outweigh the potential benefits, especially when simpler analysis or decision-making tools are sufficient to provide the desired information. In addition, developing a robust simulation study often requires specialized expertise and technical skills that may not be available in every organization or project team.

Box: Example scenario traffic flow of a large city [collapsible and expandable]

In addition, rapidly changing environments or dynamic markets may require agile decision-making, which may be hindered by complex simulation models. Ultimately, it is important to make an informed trade-off between the benefits and resources of a simulation study to ensure that the chosen method fits the solution to the specific problem.

*Please watch **video 1** carefully to understand each step and then answer the follow-up questions. **The video presentation is in german but there are english subtitles. Please make sure they are activated.***

Video 1: Simulate: yes or no? explained with an example of a food laboratory by Dr. Tatiana Starostina (03:53)

Follow-up questions to video 1

In summary, the use of simulation is particularly useful when the complexity of a system or a problem is too great for other methods. Nevertheless, it must always be checked whether the benefits of the simulation justify the effort required. This ensures that the chosen method is used efficiently and effectively in order to achieve realistic and usable results.

The advantages and disadvantages of simulation

The opportunities and challenges of simulation

Simulation offers numerous possibilities for analyzing complex issues and finding innovative solutions, but its use is not always fully advantageous. The advantages and disadvantages must be carefully weighed up in order to assess the added value of a simulation in the respective context. This chapter highlights the opportunities and challenges associated with the use of simulation methods.

The advantages of a simulation

Based on the list of reasons, the most important advantages of a simulation can be formulated:

- **Proof of function:** The functionality of a system can be tested before it exists.
- **Investigation of a complex system:** Simulation is a possible tool for finding the solution when the analytical and static method is not applicable.
- **Simplification of system analysis:** Parameters and data can be changed without having to change reality.
- **Process optimization:** The processes can be optimized according to various target functions in order to increase the productivity of the system.
- **Time-lapse system analysis:** The work of the system over a longer period of time can be simulated in a much shorter time.
- **Sensitivity analysis:** By changing individual input parameters, a sensitivity analysis can be carried out to determine the influence of the individual parameters on the system.
- **Visualization:** The processes are visualized during the simulation so that the current status of the simulation is always visible. In addition, the results of the simulation are displayed graphically (e.g. in the form of diagrams or functions).
- **Reduce costs:** Simulation can show that real system elements can be simplified or eliminated.

In today's data-driven world, simulation studies are becoming increasingly important as they help companies and organizations make informed decisions and optimize their processes in the long term.

Simulation can already be valuable in the planning phase of a project. In principle, “the earlier the better”. **Figure 2** shows the knowledge influence gap in the course of a project.

Take a look at the graphic and give it some thought.

Figure 2: Knowledge and influence dynamics in the course of the project: The role of simulation

Figure 2 shows how knowledge and opportunities to influence develop during the course of the project. At the beginning of a project, the level of knowledge is often low, but the opportunities to exert influence are high. Over time, knowledge about the system increases, while the scope for influence decreases due to decisions made and limited possibilities for adaptation. Simulations help to make well-founded decisions at an early stage by generating knowledge more quickly and reducing uncertainties. They are therefore particularly valuable in the planning phase to avoid costly wrong decisions.

The disadvantages of a simulation

In addition to the advantages, there are also the following disadvantages of a simulation:

- **Data requirements:** The basis for the analysis of a system is the system data. As a rule, numerous supply and demand data are stored in the simulator. The data must be available in good quality in order to obtain meaningful results. Sometimes not all the required data is available, so that the missing data has to be generated by the specialist knowledge or forecasts of experts, e.g. the development of traffic volumes over the next 20 years or the behavior of road users.
- **Abstraction from reality:** A simulation cannot fully represent reality. It is not easy to find the right level of abstraction for modeling a system. If the model is abstracted too much, the results of the model are not meaningful enough. On the other hand, too much detail leads to a high modeling effort. This also means that a larger number of parameters usually have to be estimated for complex models. This also requires a larger amount of data in order to obtain reliable parameter estimates. If this is not practically possible, this in turn leads to greater model uncertainties. It also makes it more difficult to adapt the model to new circumstances or a new system.
- **Time required for data collection and analysis:** Collecting and analyzing the required data is often difficult and time-consuming. In individual cases, specific surveys of the current situation may also be necessary.
- **Complex interpretation of results:** The interpretation of the results of a simulation is complicated in comparison with other methods (e.g. analytical methods).
- **Extensive expert knowledge required:** Simulation requires extensive expert knowledge, which must be acquired through training, further education and experience with simulation applications.
- **High time expenditure and high costs for model development:** Depending on the size and complexity of the system to be simulated, the time expenditure and thus also the costs for model development can be high.

The opportunities offered by simulation lie in its ability to analyze complex systems, minimize risks and support decision-making processes. Nevertheless, its use requires careful consideration, as the effort involved in obtaining data, developing models and interpreting the results can be considerable. Only if the potential

findings outweigh the associated effort and risks is the use of a simulation justified.

You can conclude this digital learning unit in the next and final chapter.

Completion of the learning unit

Answer the following questions to complete the module

Now you know when to simulate and when not to simulate and you know the advantages and disadvantages of the method.

I'm sure you can now answer a few questions.

Final learning questions

You know your way around better now.
What is an advantage and what is a disadvantage of the simulation?
Categorise the criteria correctly.
Drag them to the correct field and show your knowledge!

If you have answered both questions correctly, you will receive a badge with an award. This is visible at the top of the course. You can use it to start further self-study units online, which will be activated for you.

Enjoy!